

Product Specification

A complete technical and functional overview of Argus — architecture, monitoring engine, alerting, integrations, security, and deployment.

Overview

Argus is a self-contained network monitoring system that watches every device on a network — routers, switches, servers, firewalls, cameras, access points — and tells the right person the moment one goes down. It ships as a single Windows executable with no external dependencies: no database server to install, no account to create, no vendor cloud in the path between a device going down and someone finding out.

Everything — device inventory, historical metrics, alert history, configuration — lives in one local SQLite file next to the install. Nothing leaves the network unless explicitly configured to (an outbound webhook, an SMTP relay, a Slack webhook).

Architecture & stack

Backend: Bun + TypeScript, built on a hexagonal (ports-and-adapters) architecture — domain logic has zero dependency on any specific database, notification channel, or network library, which is what lets the same monitoring engine run identically whether backed by SQLite (the shipped Windows product) or Postgres (the multi-tenant SaaS deployment). Frontend: React + Vite + Tailwind, with Recharts/ECharts for data visualization.

Runtime	Bun (single compiled executable, no Node.js/npm required on the target machine)
Database	SQLite (embedded, exe mode) or PostgreSQL (multi-tenant SaaS mode) — same domain logic, swappable adapter
Backend framework	Hono (lightweight HTTP layer)
Frontend	React 18, Vite, Tailwind CSS, Recharts/ECharts, Framer Motion
Architecture	Hexagonal / ports-adapters — src/{domain, application, adapters, api, bootstrap}
Distribution	Single compiled .exe (portable) + an optional Inno Setup wizard installer

Core monitoring engine

Every device is monitored through one or more independent checks, each polled on its own interval:

- ICMP — ping-based reachability, latency, and packet loss.
- TCP — port-level reachability for any service.
- HTTP/HTTPS — status code, response body match, TLS validation (with a self-signed override for lab/internal appliances).
- SNMP v1/v2c/v3 — CPU, memory, interface throughput/status, and general device identity.
- Vendor REST/XML APIs — FortiGate and Sophos, see Firewall Integrations below.

A deterministic state machine (up → degraded → down → flapping, plus a maintenance override) decides device status from check outcomes — configurable consecutive-failure/consecutive-recovery thresholds avoid flapping on a single dropped ping, and a flap-detector automatically suppresses noisy alerting on a device that's rapidly bouncing between states.

ICMP is authoritative for reachability when configured: a secondary check (SNMP, HTTP, a vendor API) failing — a stale credential, a firewall rule, a misconfiguration — is recorded and visible on that check specifically, but does not by itself mark a genuinely reachable device as down.

Polling is tiered: devices in a degraded or uncertain state are polled more frequently than a stable, healthy device. Maintenance windows (one-off or recurring, scoped to a device or a group) suppress alerting for planned work without disabling monitoring itself.

Discovery & classification

Point Argus at a CIDR range and it sweeps the subnet via ARP + ICMP + TCP port probing + SNMP, then classifies each responding device using a weighted signal model: open-port fingerprint, SNMP sysDescr string, MAC-address vendor (OUI) lookup, and DHCP/reverse-DNS hostname. Each guess ships with a confidence score.

Classification isn't one-time: devices still sitting at type "unknown" are automatically re-probed on a daily schedule and reclassified once a strong signal becomes available — without ever overriding a type an admin has set manually.

Alerting & escalation

Alerts carry a severity (info / warning / critical) and a lifecycle (open → acknowledged → resolved), with a full per-alert timeline and free-text investigation notes. Tiered escalation notifies the device's owner first, then works up a configurable per-group chain if nobody acknowledges — and stops the instant someone does.

- Critical-asset instant paging — cameras and firewalls (or anything else flagged) page immediately on DOWN, skipping the storm-buffer delay and hourly rate limit.
 - Storm guard — a correlated outage is aggregated into a single storm notification instead of flooding one alert per affected device.
 - Quiet hours + severity floor, per person, with critical alerts always breaking through.
 - On-call rotations and per-group escalation chains.
 - One-click email acknowledgement links — HMAC-signed, no login required.
 - Opt-in personal digest emails (daily/weekly SLA + open-alerts summary).
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Notification channels

- Email — via the instance's own SMTP relay.
- Generic webhook — HMAC-signed for authenticity.
- Slack — severity-colored message to an Incoming Webhook.
- Microsoft Teams — MessageCard to a channel's Incoming Webhook connector, with a one-click Acknowledge action.
- PagerDuty — full Events API v2: triggers and resolves a real incident as the underlying alert opens/resolves.
- Syslog (CEF format) — for forwarding into an existing SIEM.

Firewall & vendor integrations

A dedicated Firewalls dashboard gives a fleet-wide view of every firewall device — live CPU/memory/session/VPN-tunnel metrics alongside model, firmware version, and HA role.

FortiGate — a real REST API adapter against FortiGate's monitor API: CPU, memory, active sessions, IPsec VPN tunnel status, plus identity facts (model, firmware, serial, HA role) pulled from the device.

Sophos — a REST/XML adapter against standalone on-prem Sophos Firewall's (SFOS/XG) classic management API: authenticates, validates reachability, reports IPsec VPN tunnel status. Pair with SNMP for CPU/memory.

Any other firewall vendor is fully supported through the generic SNMP checker.

SNMP support

Full SNMP v1, v2c, and v3 support (auth+priv, all standard protocols) for any SNMP-capable device.

- Bandwidth & interface monitoring — per-interface throughput and status via the standard IF-MIB, with a live interface-discovery picker.
 - CPU & memory — via the standard HOST-RESOURCES-MIB.
 - Device identity — sysDescr/sysName/sysObjectId, feeding classification and the Firewalls dashboard.
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Remote agents (multi-site / segmented networks)

A single Argus instance can only directly reach devices on networks it can route to. For a genuinely segmented environment, a lightweight standalone remote agent runs inside that segment, performs checks locally, and pushes results back to the central instance over an authenticated HTTPS connection — scoped device-by-device, flowing through the same state-machine/alerting/metrics pipeline a locally-pollled device uses.

Security

- Two-factor authentication (TOTP) with QR-code enrollment.
 - Role-based access control — admin / operator / viewer, with optional per-user group scoping.
 - Session management — every active session visible and individually revocable.
 - SSO/SAML 2.0 (hosted deployments) with auto-provisioning.
 - Scoped, read-only API keys for external integrations.
 - Full audit log — every administrative action attributed and timestamped.
 - Encrypted credential storage at rest.
 - SSRF guards on every outbound-URL-accepting feature.
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Licensing

Trial mode capped at a fixed device count, unlocked by a signed license file tied to a device-count plan. A grace period after expiry means monitoring never silently stops the moment a renewal is late.

Deployment & operations

- Runs as a native Windows service or a portable standalone executable.
- Self-heartbeat dead-man's-switch to an external push-monitor.
- One-click backup/export and restore, plus automatic scheduled backups.
- Self-update — checksum + signature verified before swapping in.
- Sub-3-second cold start; every release is smoke-tested end to end before shipping.

Dashboard & UI

- Executive Overview dashboard — fleet-wide health, availability trend, latency leaders.
- Inventory — sortable, filterable, taggable device list with bulk actions and CSV import/export.
- Live topology map — force-directed network graph with critical-asset markers.
- Bandwidth — per-interface throughput charts with the live interface-discovery picker.
- SLA & reports — downtime/availability reporting, exportable and schedulable.
- Full dark-mode support, keyboard command palette, and a kiosk mode for a NOC display.

System requirements

OS	Windows 10 / 11, 64-bit (Windows Server also supported for service deployment)
CPU / RAM	Any modern x64 CPU; 2 GB RAM minimum (scales with fleet size)
Disk	~150 MB for the application; metrics history scales with fleet size and retention
Network	Outbound access for SMTP/webhooks/update checks if used — otherwise fully offline-capable
Dependencies	None — single self-contained executable